

Stack Balancing Robot: Hybrid Controller Design – PD + CLF-QP

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Abstract

The use of robots in our day to day lives, inside the house and under a professional setting has highlighted the demand for automation, to aid in the work-life balance phenomenon. Balance is exactly the topic of concern in this paper. This project aims to develop a minimal robotic system to balance a tower of boxes equipped with a robust controller designed to oppose any wind disturbances, inspired by the Wii Party-Shifts Gifts minigame. This work focuses on modelling the environment mathematically, analysing a hybrid control system designed to keep the stack stable and balanced under external disturbances such as wind.

Keywords: *environment dynamics modelling, robot dynamics modelling, controller design, non-linear control systems, linear control systems, underactuated systems.*

1. INTRODUCTION

Taking inspiration from a simple physics based minigame I grew up playing, I decided to embark on this journey to replicate the same game physics but make a robot do it or balance the stack. This minigame involved balancing a tower of gift boxes against external disturbances like wind acting along one direction. The aim of the game is to balance the boxes or hold the tower upright without toppling it.

However, not a lot of research went into this aspect of balance. Balance in terms of robotics means how well the robot can navigate the terrain

without falling over and this is called self-balance like the ballbot [1]. This is the research field that accommodates a lot of researchers which overshadowed the other side of the coin – balancing an object on a robot.

Therefore, the problem statement can be defined as balancing a tower of boxes and maintaining the upright position. This project will introduce a rectangular base plate hinged on its centre about its y-axis. This plate will be tasked with balancing N cube-shaped boxes of different sizes stacked on top of each other from the biggest being in contact with the base plate to smallest at the top if the stack.

This report will cover the dynamics modelling of the environment and robot, stability analysis, controller architecture, simulation and results.

2. DYNAMICS MODELLING

The dynamics of the system is modelled on the Lagrange Equations of motion (LHS) along with the non-conservative forces (RHS) [2].

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = B\tau + J(q)^T\lambda + F_{wind}(q, \dot{q}) - D\dot{q}, \quad (1)$$

$M(q)$ = Inertia Matrix,

$C(q, \dot{q})\dot{q}$ = Coriolis & Centrifugal Forces,

$G(q)$ = Gravity Vector,

$J(q)^T \lambda = \text{Contact Forces},$

$\tau = \text{Motor torque at hinge},$

$B = [1 \ 0 \ 0 \ \dots \ 0]^T, \text{ because only actuated at } \theta_0$

$F_{wind}(q, \dot{q}) = \frac{1}{2} \sum_{i=1}^N \rho C_d A_i v^2$

$D = \text{Viscous damping of the base plate}$

$[q, \dot{q}]^T = x, \text{ state space variable}$

$\text{size}([q, \dot{q}]^T) = 1 + 2N, N = \# \text{ of boxes}$

2.1. State Space Variable Definition:

The state space variable (x) can be defined as a column vector including:

- Pitch angle of the plate (θ_0),
- Slip distance of each box (y_i) – difference between the plate centre of mass (CoM) and box's CoM,
- Pitch angle speed of the plate ($\dot{\theta}_0$),
- Slip speed of each box ($\dot{y}_i$)

$$q = [\theta_0, y_1, y_2, y_3, \dots, y_N, \theta_1, \theta_2, \theta_3, \dots, \theta_N]^T$$

$$\dot{q} = [\dot{\theta}_0, \dot{y}_1, \dot{y}_2, \dot{y}_3, \dots, \dot{y}_N, \dot{\theta}_1, \dot{\theta}_2, \dot{\theta}_3, \dots, \dot{\theta}_N]^T$$

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$$

2.2. Complementarity Conditions:

This subsection will introduce friction dynamics modelled under 2 situations: Stick and Slip as shown in Eq (2) and Eq(3) respectively.

Friction modelling is important because it directly affects the $J(q)^T \lambda$ which corresponds to the contact forces/torques term.

$$\lambda_{t,i} = -F_{net,i}, \text{ if } |\lambda_{t,i}| \leq \mu_s \lambda_{n,i} - \text{Stick (2)}$$

$$\lambda_{t,i} = \mu_k \lambda_{n,i} \text{sign}(\dot{y}_i), \text{ if } |\lambda_{t,i}| > \mu_s \lambda_{n,i} - \text{Slip (3)}$$

$\mu_s = \text{static friction coefficient}$

$\mu_k = \text{kinetic friction coefficient}$

Stick situation – A situation in which all boxes are stuck to each other and the base of the stack is stuck to the plate. This simply means that there is no kinetic friction, and static friction is responsible for holding all of these in place. Since static friction is the only friction in play, $J(q)^T \lambda \rightarrow 0$ as $\lambda \rightarrow 0$ because the static friction produces contact forces that oppose the internal energy of the boxes themselves and hence contributes a net-zero towards the external dynamics of the system. A situation in which $y_i = \dot{y}_i = 0$.

Slip situation - A situation in which one or more boxes' CoM is not aligned to the CoM axis of the plate's hinge joint. This means that $y_i \neq 0$, therefore the box is slipping from the stack which means that there is a kinetic friction in play. This friction will oppose the wind force and/or the force induced by the passive dynamics of the system alongside the torque input.

Putting the modelled dynamics together, a control affine form as shown in Eq (4) can be obtained.

$$\dot{x} = f(x) + g(x)\tau + w(x) \quad (4)$$

Where, $f(x)$ is the drift caused by passive dynamics of the system corresponding to the plate inertia ($M(q)$), Coriolis/centrifugal force ($C(q, \dot{q})\dot{q}$), gravity ($G(q)$), and contact forces ($J(q)^T \lambda$), $g(x)$ is the column vector that activates the actuator and $w(x)$ is the wind force ($F_{wind}(q, \dot{q})$), $w(0) = 0$

$$\dot{x} = \begin{bmatrix} \dot{q} \\ \ddot{q} \end{bmatrix} = \begin{bmatrix} \dot{q} \\ \frac{B\tau + J(q)^T \lambda + F_{wind}(q, \dot{q}) - D\dot{q} - C(q, \dot{q})\dot{q} - G(q)}{M(q)} \end{bmatrix} \quad (5)$$

The objective is to obtain a torque such that $x = [0, 0]$ or the system stays upright. To maintain an upright position, the system needs to stabilize at that position. The next section will focus on analysing the stability of the upright position based on the modelled environment and robot dynamics.

3. STABILITY ANALYSIS

The robot is required to balance the system or bring it to an upright position given any state; therefore, the system must be globally asymptotically stable about the upright position or at origin. In this section, the approach taken to achieve global asymptotic stability at the origin will be described.

To achieve global asymptotic stability at the origin, the origin must be a stable equilibrium point, and all trajectories should converge to this equilibrium point. This section will be divided into two subsections: Equilibrium Point Analysis, and Global Asymptotic Stability Analysis.

3.1. Equilibrium Point Analysis

This section will focus on proving that the origin is indeed an equilibrium point setting it up for further stability analysis.

$$\text{Origin: } x = 0, \text{ i. e. } q = 0, \dot{q} = 0$$

Definition -

The origin is an equilibrium point iff,

$$\dot{x} = 0, \text{ i. e. } \dot{q} = 0 \text{ \& } \ddot{q} = 0$$

Proof –

By default, $\dot{q} = 0$, because $x = 0$. For $\ddot{q} = 0$, $B\tau + J(q)^T \lambda - G(q) = 0$, based on Eq (1). This means that there are forces due to uncontrolled dynamics always acting on the system. Therefore, a holding torque as a feedforward term just enough to oppose the dynamics' inherent torque ($-J(q)^T \lambda + G(q)$) should be added to the system to maintain the upright position to avoid tilting or toppling of the stack even without external disturbances.

Conclusion -

Therefore, given that $B\tau + J(q)^T \lambda - G(q) = 0$, origin is equilibrium.

3.2. Global Asymptotic Stability Analysis

This subsection deals with the stability analysis of the equilibrium point under analysis. The Lyapunov Global Asymptotic Stability theorem will be leveraged in this subsection to prove that the origin is a globally asymptotically stable point.

Definition -

Equilibrium point is globally asymptotically stable iff there exists a function V (a Lyapunov candidate function) such that:

$$V(0) = 0$$

$$V(x) > 0, |x| \rightarrow \infty$$

(a global positive definite function)

$$\dot{V}(x) < 0$$

Proof -

Let $V(x)$ equal to the energy function:

$$V(x) = KE_{tot} + PE_{tot} \quad (6)$$

$$KE_{tot} = KE_{plate} + KE_{boxes}$$

$$\therefore KE_{tot} = \frac{1}{2} J_o \dot{\theta}_0^2 + \frac{1}{2} \sum_{i=1}^N (m_i (\dot{y}_i + h_i \dot{\theta}_0)^2 + I_i \dot{\theta}_i^2)$$

Where I_i is the moment of inertia of box i

$$PE_{tot} = PE_{plate} + PE_{boxes},$$

$$PE_{plate} = 0,$$

because hinge is on CoM of plate

$$\therefore PE_{tot} = \sum_{i=1}^N m_i g h_i (1 - \cos(\theta_0 + \theta_i))$$

$$\therefore V(x) = \frac{1}{2} J_o \dot{\theta}_0^2 + \frac{1}{2} \sum_{i=1}^N (m_i (\dot{y}_i + h_i \dot{\theta}_0)^2 + I_i \dot{\theta}_i^2) + \sum_{i=1}^N m_i g h_i (1 - \cos(\theta_0 + \theta_i)) \quad (7)$$

Prove $V(0) = 0$:

$$\begin{aligned} x = 0 &\rightarrow q = 0 \text{ \& } \dot{q} = 0 \\ &\rightarrow \theta_0 = 0, \theta_i = 0, y_i = 0 \\ &\rightarrow \dot{\theta}_0 = 0, \dot{\theta}_i = 0, \dot{y}_i = 0 \\ \therefore V(0) &= 0 \end{aligned}$$

Prove $V(x) > 0$:

$$\frac{1}{2} J_0 \dot{\theta}_0^2 > 0 \text{ for all } x, |x| \rightarrow \infty \text{ \& } x \neq 0,$$

$$\frac{1}{2} \sum_{i=1}^N \left(m_i (\dot{y}_i + h_i \dot{\theta}_0)^2 + I_i \dot{\theta}_i^2 \right) > 0,$$

$$\text{for all } x, |x| \rightarrow \infty \text{ \& } x \neq 0,$$

This is because of the squared terms.

$$\sum_{i=1}^N m_i g h_i (1 - \cos(\theta_0 + \theta_i)) > 0,$$

$$\text{for all } x, |x| \rightarrow \infty \text{ \& } x \neq 0,$$

$$\text{because } 1 - \cos(\theta_0 + \theta_i) > 0 \text{ for all } \theta_0 < 90^\circ, \theta_0 \neq 0 \text{ \& } \theta_i < 90^\circ, \theta_i \neq 0$$

$$\therefore V(x) > 0$$

Prove $\dot{V}(x) < 0$:

$$\begin{aligned} \dot{V}(x) = J_0 \dot{\theta}_0 \ddot{\theta}_0 &+ \sum_{i=1}^N (m_i (\dot{y}_i + h_i \dot{\theta}_0) (\ddot{y}_i + h_i \ddot{\theta}_0) + I_i \dot{\theta}_i \ddot{\theta}_i) \\ &+ \sum_{i=1}^N m_i g h_i (\dot{\theta}_0 + \dot{\theta}_i) \sin(\theta_0 + \theta_i) \end{aligned} \quad (8)$$

The above function (Eq (8)) must be a negative definite function for the equilibrium point to be globally asymptotically stable however there is no concrete proof that the above function is negative definite because the polarity depends on the state variables ($\dot{\theta}_0, \dot{y}_i$ \& $\dot{\theta}_i$) that are positive or negative definite therefore, it is impossible to deduce the polarity of each term as each state variable could be of different polarities. However, introducing a bound to this function will overcome this problem and prove the statement.

$$\dot{V}(x) \leq -\alpha V(x) < 0 \rightarrow QP \text{ Condition}$$

This condition states that there exists an $\alpha > 0$ such that $-\alpha V(x) < 0$. Since $-\alpha V(x)$ is negative definite and $\dot{V}(x) \leq -\alpha V(x)$ then $\dot{V}(x)$ is negative definite too.

Conclusion -

Hence, proves asymptotic stability is achievable at origin.

4. CONTROLLER ARCHITECTURE

This section discusses the controller design and architecture to achieve the task in hand. We will split this section into 3 subsections – Feed-Forward Control, Linear Control \& Non-Linear control. These three sections together will introduce the hybrid control system proposed to complete objective.

4.1. Feed-Forward Control

The section will discuss the design of a controller that will produce the holding torque to oppose the inherent torque produced by the dynamics of the system.

The objective of this controller is to eradicate any forces/factors opposing the stability of the plate and tower in the upright position. The passive dynamics causes the torque because of the inertial and Coriolis terms of the plate and system, as a result driving the plate to tilt toward one direction due to the imbalance or very negligible slip distance between the blocks and the plate.

Accounting for the passive dynamics in play and applying a resistive torque allows the plate to keep the tower in an upright position and maintains the plate's initial pitch angle at 0° . This torque is called the holding torque. This holding torque will allow for the origin to be a stable equilibrium point as discussed in Section 3.1

$$B\tau + J(q)^T \lambda - G(q) = 0$$

But $\lambda = 0 \rightarrow J(q)^T \lambda = 0$ because it is assumed that it is a stick-only system (Refer to Section 2.2). Therefore, the torque input/holding torque should maintain upright position under an open loop setting by opposing the static gravity torque/the passive dynamics.

$$B\tau = \tau_{ff} = -G(q) = -\sum_{i=1}^N m_i g h_i \sin(\theta_0 + \theta_i) \quad (9)$$

4.2. Linear Control

This section will discuss the architecture of the linear controller responsible for maintaining the stability of the system for small perturbations defined within the region of attraction (RoA). This linear controller makes use of the PD control system that outputs a torque value based on the error or the perturbation, in this case, to stabilize the system under small angular shifts, obtained by linearizing the control-affine form about the origin.

$$e(error) = \theta_{ref} - \theta_0 = -\theta_0, \text{ as } \theta_{ref} = 0$$

$$\tau_{PD} = -K_p \theta_0 - K_d \dot{\theta}_0 \quad (10)$$

4.3. Non-linear Control

This subsection will introduce the nonlinear controller that will accompany the linear controller discussed in the previous subsection to maintain stability of the system for large perturbations defined outside the RoA. This nonlinear controller makes use of the CLF-QP control system law and bounds the time derivative of the energy function such that asymptotic stability can be achieved. This makes use of Lyapunov Stability Theorems, CLF-QP laws and dynamics model of the system to derive the right torque value required to oppose the forces impacting the stability of the system. [3,4]

Eq (1) can be rewritten to state specific formulae as shown below:

$$\frac{d}{dt} \left(\frac{\partial KE_{tot}}{\partial \dot{q}_j} \right) - \frac{\partial KE_{tot}}{\partial q_j} + \frac{\partial PE_{tot}}{\partial q_j} = Q_j \quad (11)$$

Where Q_j represents the net non-conservative forces acting on that state variable q_j .

This breakdown provides a more detailed understanding of the entire system in terms of each state variable. This breakdown is required to formulate the CLF-QP equation to determine the torque required to maintain stability at origin under external disturbances like wind. The CLF-QP equation will be formulated by manipulating Eq (7) and (11) and the QP condition introduced in Section 3.2

Plate ($q_j = \theta_0$):

$$J_0 \ddot{\theta}_0 + \sum_{i=1}^N m_i h_i (\ddot{y}_i + h_i \ddot{\theta}_0) + \sum_{i=1}^N m_i g h_i \sin(\theta_0 + \theta_i) = \tau + J(\theta_0)^T \lambda + F_{wind}(\theta_0, \dot{\theta}_0) - D \dot{\theta}_0 \quad (12)$$

Box Slip ($q_j = y_i$):

$$m_i (\ddot{y}_i + h_i \ddot{\theta}_0) = J(y_i)^T \lambda + F_{wind,i}(y_i, \dot{y}_i) - D \dot{y}_i \quad (13)$$

Box Pitch ($q_j = \theta_i$):

$$I_i \ddot{\theta}_i + m_i g h_i \sin(\theta_0 + \theta_i) = J(\theta_i)^T \lambda + F_{wind}(\theta_i, \dot{\theta}_i) - D \dot{\theta}_i \quad (14)$$

Further simplification of this leads to,

$$I_i \ddot{\theta}_i = \tau_{contact}^{(i)} + \tau_{wind}^{(i)} - D_{\theta,i} \dot{\theta}_i \quad (15)$$

Putting Eq (12), (13) and (15) together along with the QP condition, the general form CLF-QP shown below can be rewritten to find the torque required to converge to origin.

General Form CLF-QP:

$$\dot{V}(x) = L_f V(x) + L_g V(x) \tau \leq -\alpha V(x) \quad (16)$$

$$L_g V(x) = \frac{\partial V}{\partial \dot{\theta}_0} \frac{\partial \ddot{\theta}_0}{\partial \tau} = \frac{\left(\frac{dV}{d\dot{\theta}_0} \right)}{J_0} \equiv \frac{\beta(x)}{J_0},$$

$$L_f V(x) = \frac{dV}{d\dot{\theta}_0} (-D \dot{\theta}_0 - \sum_i \lambda_{t,i} h_i) \equiv \beta(x) \text{drift}(x)$$

$$\sum_i \lambda_{t,i} h_i = \text{contact torque}$$

$$\frac{\beta(x)}{J_0}\tau \leq -\alpha V(x) - \beta(x)\text{drift}(x) \rightarrow QP - \text{Constraint}$$

$$\therefore \tau_{clf} = -\frac{\alpha V(x) + \beta(x)\text{drift}(x)}{(\beta(x)/J_0)}, (17)$$

$$-\tau_{max} \leq \tau_{clf} \leq \tau_{max}, \alpha > 0$$

Therefore, to control the system effectively,

$$\tau = \tau_{ff} + \tau_{PD} + \tau_{clf}$$

$$\dot{x} = f(x) + g(x)\tau + w(x)$$

5. SIMULATION

The world was setup in Unity3D using its built-in physics engine to determine the contact torque as shown below.

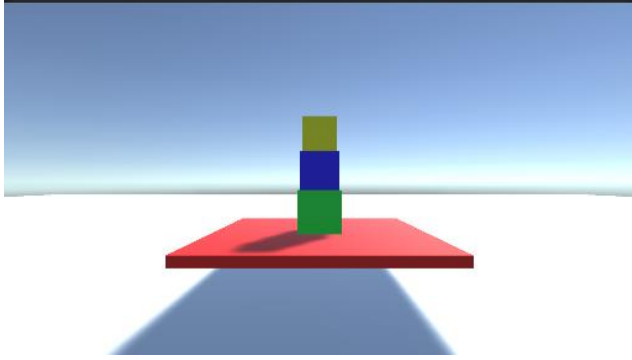


Fig1. Unity3D Environment

Fig1. shows three boxes of varying sizes and masses stacked on top of a rectangular plate that rotates about the y-axis, pivoted at its CoM. The environment consists of 3 boxes, meaning $N = 3$, where box 1 is box at the base of the stack (green) and box 3 is the box at the top of the stack (yellow). The parameters and constants were defined as below:

$$m_{plate} = 3kg,$$

$$D = 0.2, J_0 = 0.063, C_d = 1.2, \rho = 1.225,$$

$$m_1 = 5, m_2 = 3, m_3 = 1, \tau_{max} = 2,$$

$$h_1 = 0.06, h_2 = 0.155, h_3 = 0.24,$$

$$\text{small perturbations} = 0.3^\circ,$$

$$K_d = 20, K_p = 200, \alpha = 100,$$

$$\mu_{s,i} = 0.05, \mu_{k,i} = 0.2, \text{ where } i = 1, 2, 3$$

CLF-QP control activates after perturbation (i.e. error = $-\theta_0$) is bigger than 0.3° . This value was chosen by trial and error based on the complementarity of the friction functions discussed in Section 2.2

Wind was applied at a steady increasing rate:

$$v = v(t) = v_{step}1_{t>t_0} + A_{sin}\sin(2\pi ft),$$

$$t_0 = 2, A_{sin} = \text{Sine Amplifier} = 1.5,$$

$$f = \text{Sine Frequency} = 0.4, v_{step} = 12$$

The box-plate and box-box contact torques are read real-time using the software's inbuilt contact force API

6. RESULTS

As mentioned in the previous section,

$$K_p = 200, K_d = 20, \alpha = 100$$

These gain constants and decay value was obtained after fine tuning as much as possible. Even after fine tuning, the system oscillates about the $\theta_0 = 0$ axis without convergence. The wind speed $v(t)$ ramps linearly from 0 to 12 m/s at a rate of 0.3m/s from $t = 2$ s as shown in Fig2, generating a wind force that grows over time. The PD controller manages to maintain the system in the upright position until $t \sim 40$ s and the CLF controller seems to try to realign the plate as soon as its activated but fails to do so due to the actuator limitations and simulation limitations as shown in Fig3.

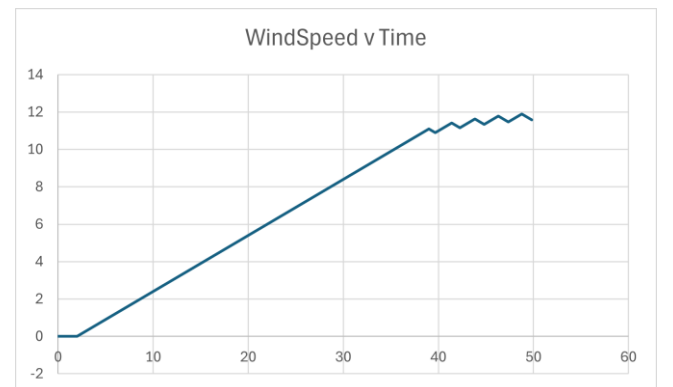


Fig2. Windspeed vs Time

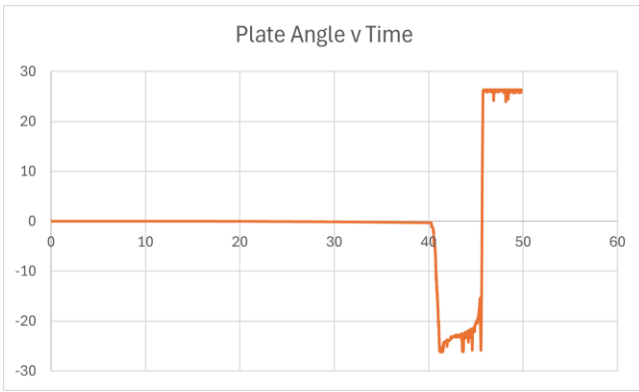


Fig3. Plate Angle vs Time

Limitations:

High computational load limited time step and integration accuracy of the simulator. The full drag model for computing wind force is applied at each box's centre of mass rather than a distributed force over the face. The wind is treated as a planar force throughout this project as it acts only in the x-direction. However, wind applies force from all directions. Unrealistic gain constants and decay values. $K_p = 200$ is far larger than any real actuator can sustain at the plate's inertia. Assumes that the gravity torque is always nullified by the feed forward torque, but external forces may add on to the static torques.

7. CONCLUSION & FUTURE SCOPE

Achieving balance is, therefore, theoretically possible however replicating it in the real world can be difficult or near impossible with current actuator limits.

This work can be extended further by implementing adaptive CLF, sliding-mode controllers or reinforcement learning paradigms to obtain realistic, fine-tuned parameters to run on simulation and implement a Sim2Real pipeline to develop a robotic system that can balance the stack. The addition of extra DoFs (Degrees of Freedom) to the balancing plate would allow for more control over the movements, for example, 4 prismatic joints on the corners of the plate allow for more control over the system and it will be able to account for wind forces from all directions and create a more robust real-time control system.

8. REFERENCES

1. Nagarajan U, Kantor G, Hollis R. The ballbot: An omnidirectional balancing mobile robot. *The International Journal of Robotics Research*. 2013;33(6):917-930. doi:10.1177/0278364913509126
2. Y. Wang, N. Dehio and A. Kheddar, "On Inverse Inertia Matrix and Contact-Force Model for Robotic Manipulators at Normal Impacts," in *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 3648-3655, April 2022, doi: 10.1109/LRA.2022.3145967.
3. A. D. Ames, K. Galloway, K. Sreenath and J. W. Grizzle, "Rapidly Exponentially Stabilizing Control Lyapunov Functions and Hybrid Zero Dynamics," in *IEEE Transactions on Automatic Control*, vol. 59, no. 4, pp. 876-891, April 2014, doi: 10.1109/TAC.2014.2299335.
4. A. D. Ames, J. W. Grizzle and P. Tabuada, "Control barrier function based quadratic programs with application to adaptive cruise control," *53rd IEEE Conference on Decision and Control*, Los Angeles, CA, USA, 2014, pp. 6271-6278, doi: 10.1109/CDC.2014.7040372.